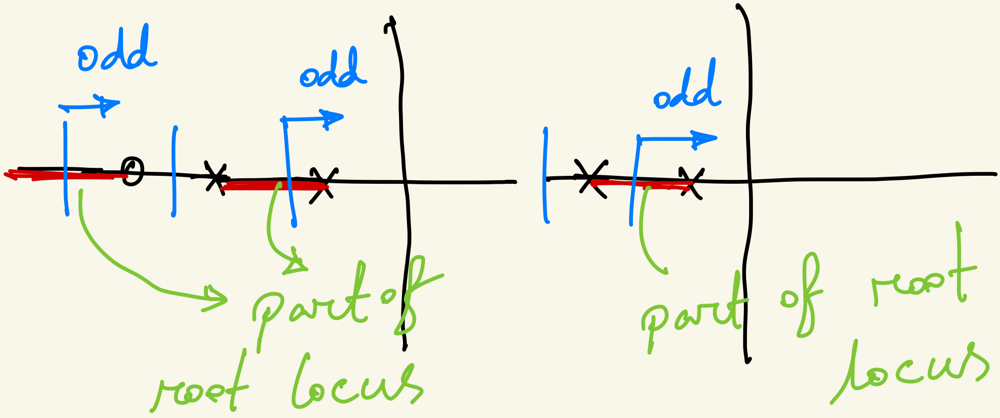
A few rules for plotting root locus :-

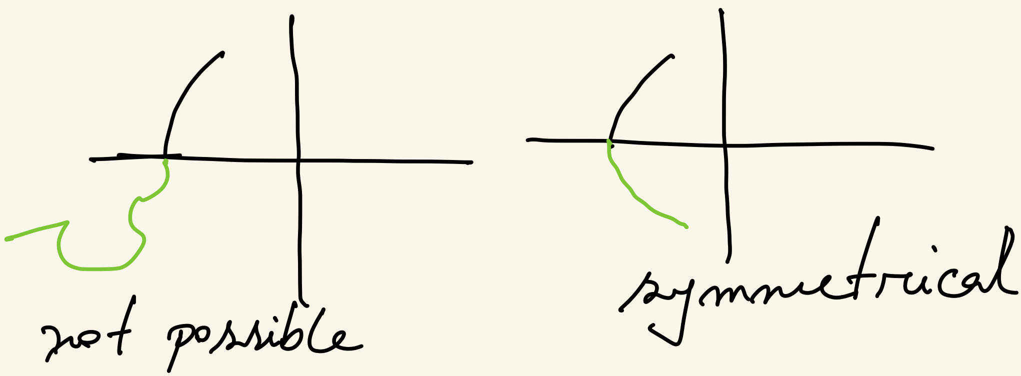
Rule 1 : The n branches of the locus start at the poles of $L(s)$ and m of these branches end on the zeros of $L(s)$.

n = number of poles
m = number of zeros
$n-m$ = number of asymptotes

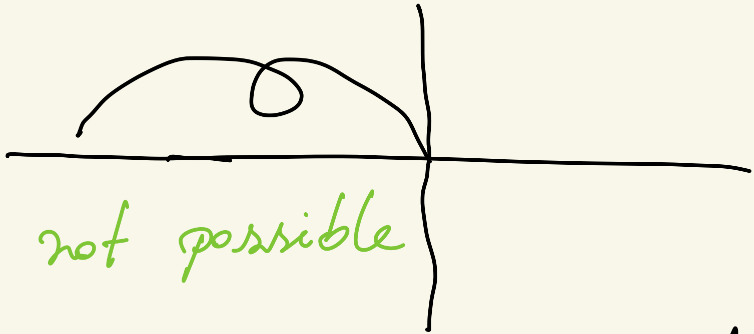
Rule 2 : The loci are on the real axis to the left of an odd number of poles and zeros.



Rule 3: The root loci must be symmetrical with respect to the horizontal axis.



Rule 4: The root loci cannot cross its pat.



not possible

Rule 5 : For large s and K, $n-m$ branches of the loci are asymptotic to lines at angles φ_l radiating out from the point $s=\alpha$ on the real axis, where

$$\varphi_l = \frac{2l+1}{n-m} 180^\circ, \quad l = 0, 1, 2 \cdots$$

$$s = \alpha = \frac{\sum P_i - \sum z_i}{n-m}$$

centroid of the asymptote

There are several other rules. However, with the above mentioned 5 rules, we will be able to draw most of the root locus.

A few examples:-

$$L(s) = \frac{1}{s(s+1)}$$

$n = 2, \quad m = 0$

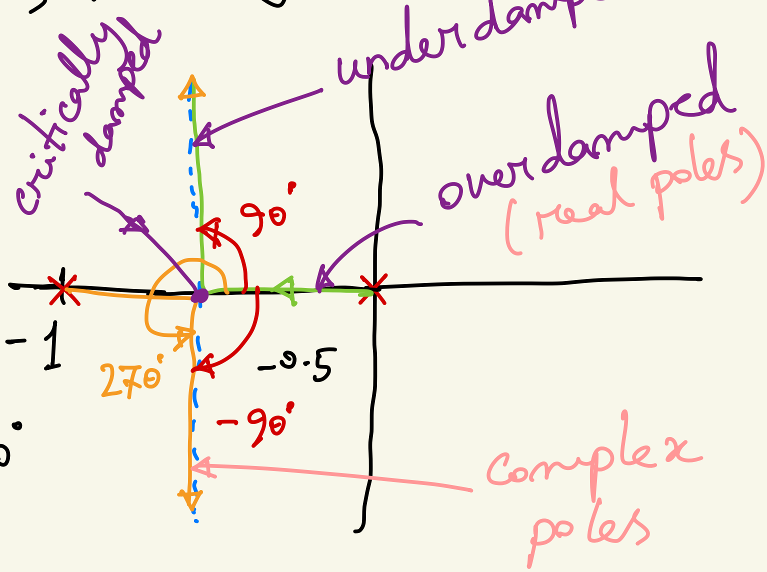
$n - m = 2$, two asymtotes

$$\varphi_1 = \frac{2l+1}{n-m} \cdot 180^\circ$$

$$= \frac{2 \cdot (0) + 1}{2} \cdot 180$$

$$= 90^\circ$$

$$\varphi_2 = \frac{2(1)+1}{2} \cdot 180^\circ$$



$= \frac{3}{2} \times 180^\circ$

$= 270^\circ$

$= -90^\circ$

Centroid

$\sigma = \alpha = \frac{\sum P_i - \sum Z_i}{n - m}$

$= \frac{(0 - 1) - (0)}{2}$

$= -0.5$